EMOTIONAL ARCHITECTURE

with A.I.

CAN A.I. UNDERSTAND HUMAN EMOTIONS?

A Computational Side Project on Emotion & Geometry

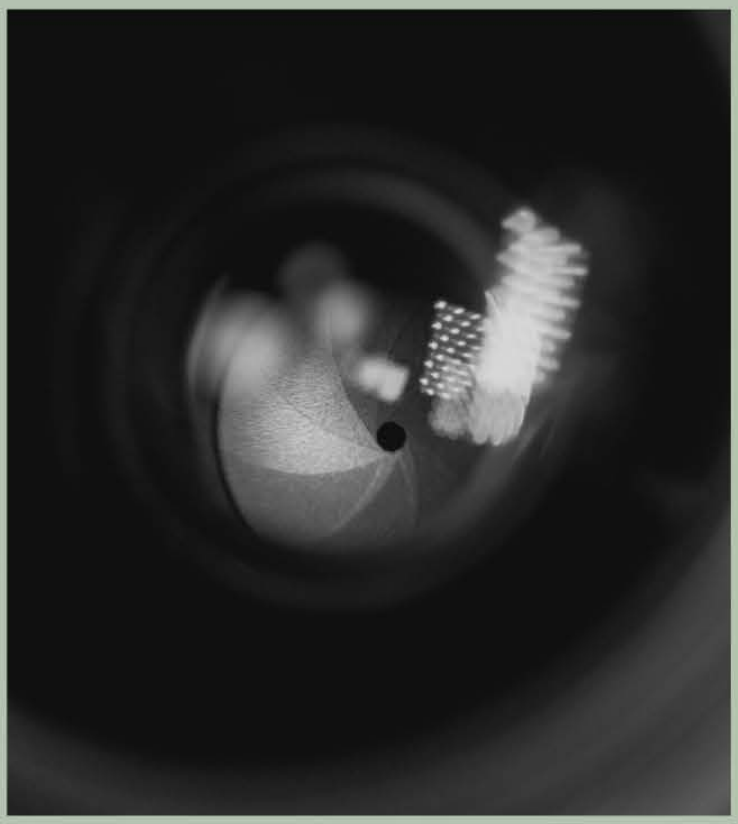
DATA Facial Expression
EEG Neural Signals

ALGORITHM Generative AI
Computer Vision
Rhino Python

VISUALIZATION Three.js
ComfyUI

01. 2026.

CONTENTS

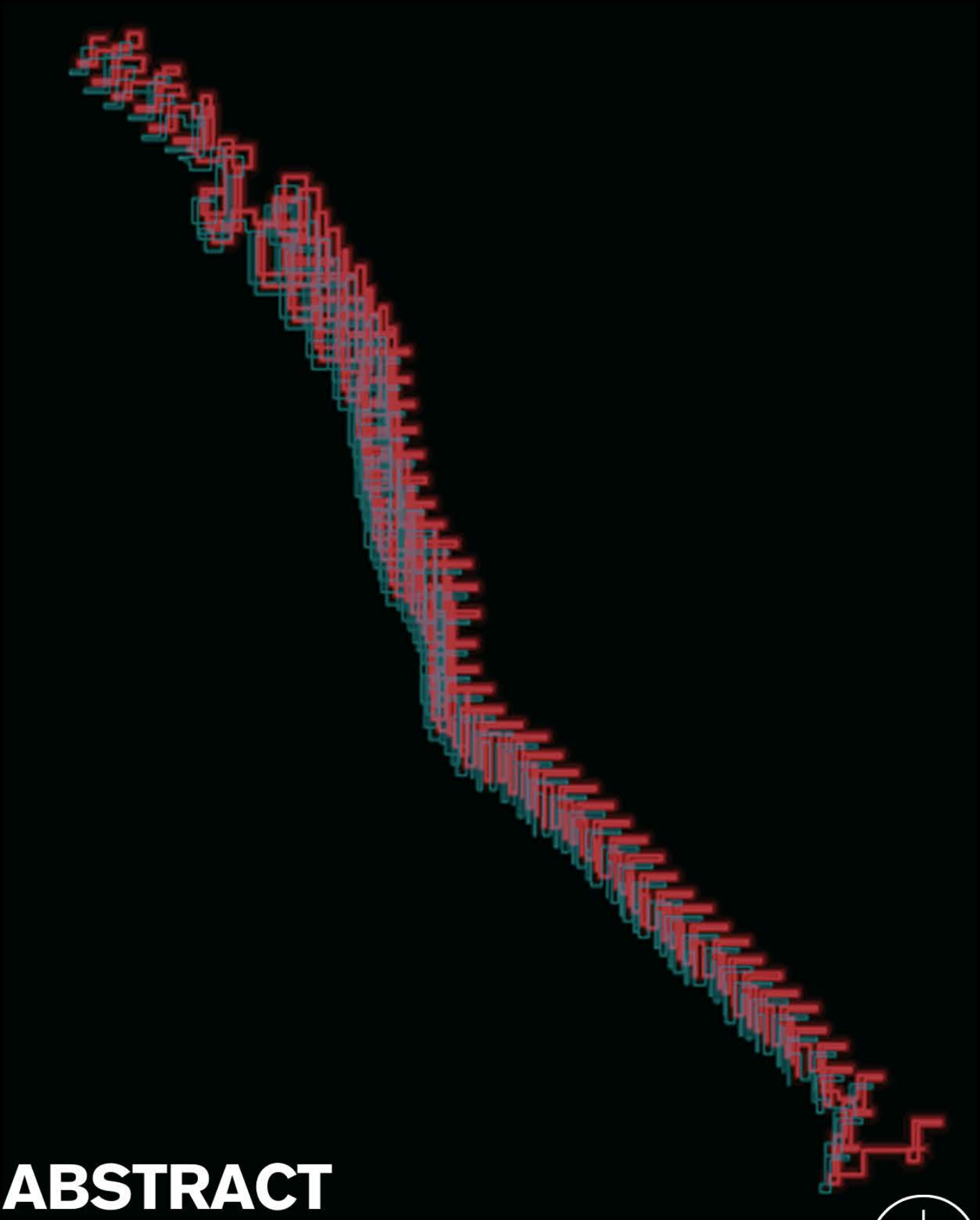


03 **ABSTRACT**

07 **RE-IMPLEMENTATION**

11 **SPATIALIZATION**

15 **CONCLUSION**



ABSTRACT

BACKGROUND & CONTEXT:
THE EMOTIONAL DISTORTION OF SOCIETY

This project emerges as an experimental side project running parallel to my ongoing graduation project, 'Ground 0'. While the project explores the spatial relationship between affect and urban space, this experiment shifts the focus to immediate interaction and computational visualization.

Before diving into the main body of the research, this section introduces the background and motivation behind the graduation project.

Emotion acts as a layered system(Fig. 01)—originating from sensory stimuli, processed by consciousness, and expanded through social interaction. However, modern society distorts this fundamental system(Fig. 02). In the age of social media, emotional expression is restricted to limited windows, and feelings are commodified. We are losing the ability to reflect on our own genuine emotions, trapped in a cycle where affect is manufactured rather than felt.

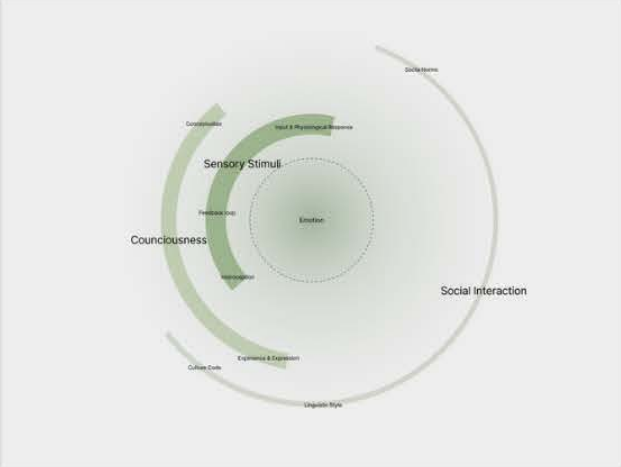


Fig. 01 layered system of Emotion

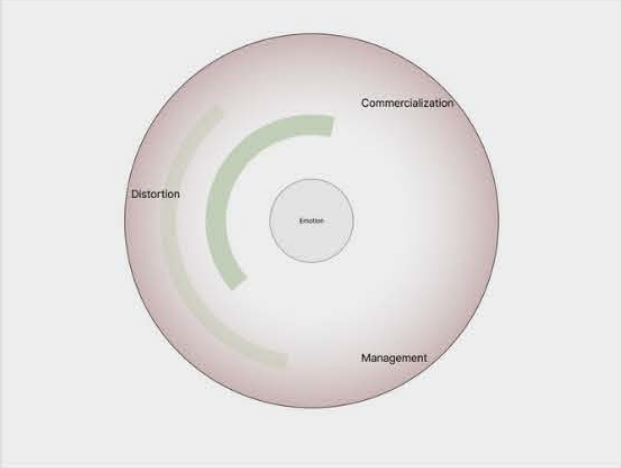


Fig. 02 Social Influence on Emotions

DO WE TRULY UNDERSTAND OUR OWN EMOTIONS?

Architectural attempts to address this issue have often fallen short. Neuroarchitecture rely too heavily on visual hierarchy, offering a fragmented understanding of human affect. Even Architectural Phenomenology, which seeks to relate the individual to the world, risks generalizing sensory experience. Instead of allowing raw feelings to emerge, it often attempts to dictate specific emotions within a space, creating a "curated" atmosphere rather than a genuine one.

Fig. 03 NeuroArchitecture Experiment

In an era where genuine emotion is fading, my goal is to reawaken the dulled senses. I aim to create a 'field' where the sensory system is restored, allowing humans to feel the world again without distortion.

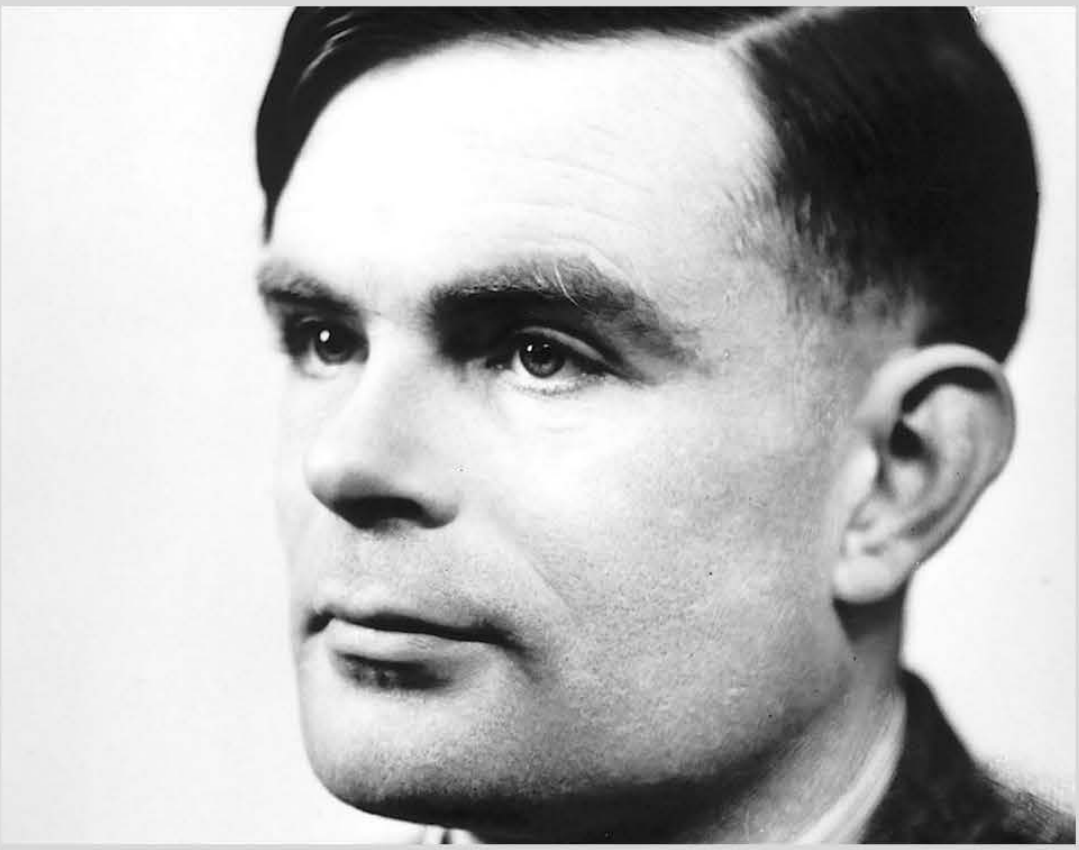
Fig. 04 Therme Vals, Peter Zumthor

OR DO WE EVEN KNOW HOW?

Ironically, I thought Artificial Intelligence serves as the ideal partner for analyzing human affect.
This project began with a hypothesis: perhaps AI, which aggregates the average values of countless individuals, is uniquely positioned to capture the essence of 'universal human emotion'—free from subjective distortion.

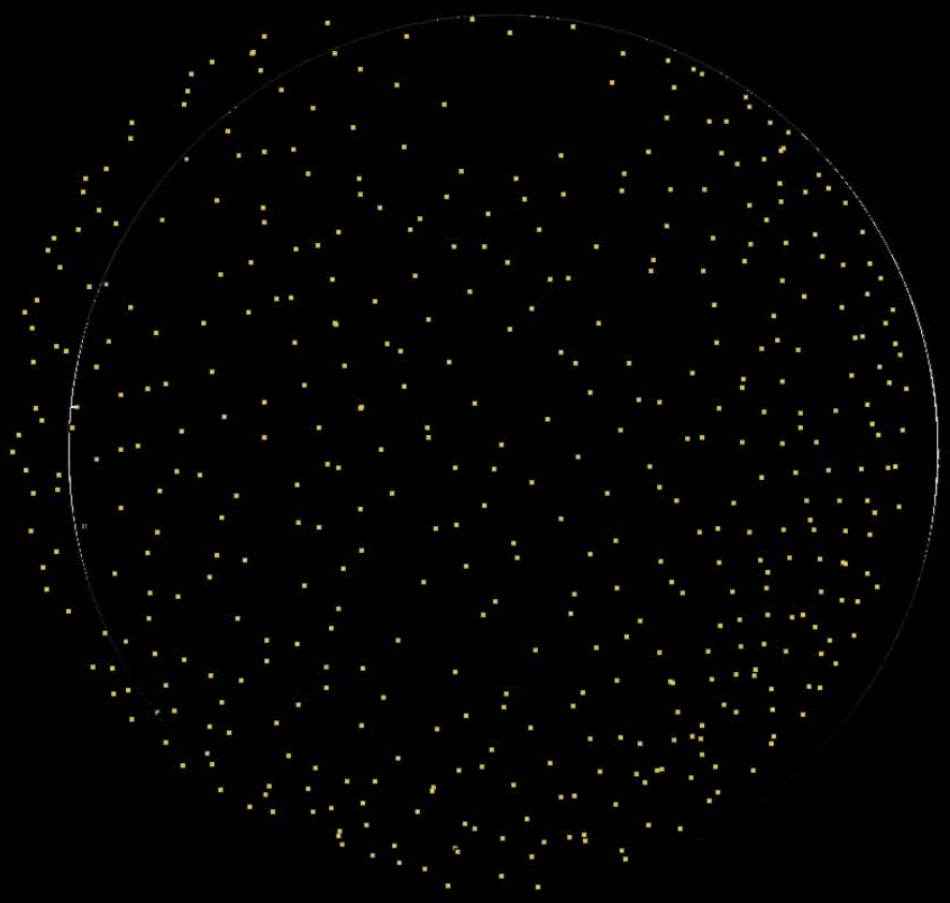
"It is customary to offer a grain of comfort, in the form of a statement that some peculiarly human characteristic could never be imitated by a machine.

**I cannot offer any such comfort,
for I believe that no such bounds can be set."**



Alan Turing, broadcast on BBC Third Programme (1951)

RE
IMPLEMENTAION



EMOTIONAL SPHERE

"Can code capture the warmth of a smile? If so, How do we observe?"

This project starts with a simple curiosity: "If we ask AI to express human emotion, can we still feel empathy toward it?"

We interact with AI every day, but we rarely see how it sees us. Does it understand our joy, or does it just calculate vectors?

AI possesses no emotion. Yet, it is trained on billions of human data points. Paradoxically, the output generated by AI might represent the most 'universal average' of human affect.

Inquiring with both Gemini and GPT on how to re-implement human emotion. Among the suggestions, I adopted the method that was the most intuitive, yet avoided the literal depiction of facial expressions. (by GPT 5.1)

```
> [SYSTEM START] Vision API Active...
> TARGET IDENTIFIED : Subject_01
> TIMESTAMP : 2026-01-10 14:45:22

[RAW METRICS DETECTED]
- Emotion_Joy : "VERY_LIKELY" (Score: 0.95)
- Emotion_Sorrow : "VERY_UNLIKELY" (Score: 0.01)
- Head_Pan_Angle : -12.5° (Left_Tilt)
- Head_Roll_Angle : +5.2° (Upward)
- Landmark_3D : {x: 245, y: 312, z: -40}
```

For step 3, Original Python code is withheld for cloud security. Replaced with a simplified logic.

01. EMOTION SCALING

Function: likelihood_to_score()

Mapping Table:

- VERY_UNLIKELY 0.0
- POSSIBLE 0.5
- VERY_LIKELY 1.0

02. SPATIAL RE-MAPPING

Function: normalize_angle()

- Converting raw head-pose degrees (Pan/Tilt/Roll) into a normalized vector space (-1.0 ~ 1.0).

WORKFLOW

STEP 1: INPUT

- Webcam Feed : capturing live facial expressions with webcam.

STEP 2: ANALYSIS

- Google Vision API: detecting facial landmarks and calculating emotion probability vectors.

STEP 3: TRANSLATION

- Python & Flask Server: normalizing raw data values and converting them into JSON format for real-time streaming.

STEP 4: VISUALIZATION

- Three.js: rendering a dynamic particle sphere where vertex displacement is driven by emotional intensity.

03. RESPONSE STRUCTURE

Status : 200 OK (Latency < 100ms)

Face Data Object:

- Emotions: { joy, sorrow, anger, surprise }
- Pose: { pan, tilt, roll, box_size }
- Rect: { min_x, min_y, max_x, max_y }

04. CLOUD ARCHIVING

Target: Google Cloud Storage (GCS)

Format:

- raw_image_{uuid}.jpg (Original Frame)
- analysis_{uuid}.json (Vector Data)

EXPERIENCE GUIDE

COLLECTED DATA COULD BE USED FOR FURTHER RESEARCH
수집된 데이터는 추후 연구를 위해 활용될 수 있음

TRY NOW

1. Look at the tablet next to this booklet.
2. Start and Make a face.
3. See how the AI interprets you.
4. Collect 'Emotions' as much as you can
5. Or you can use your own device

You can see whole HTML codes for visualization via Github

SPATIALIZATION

EMOTIONAL SPACE

Codes, Source, Workflow

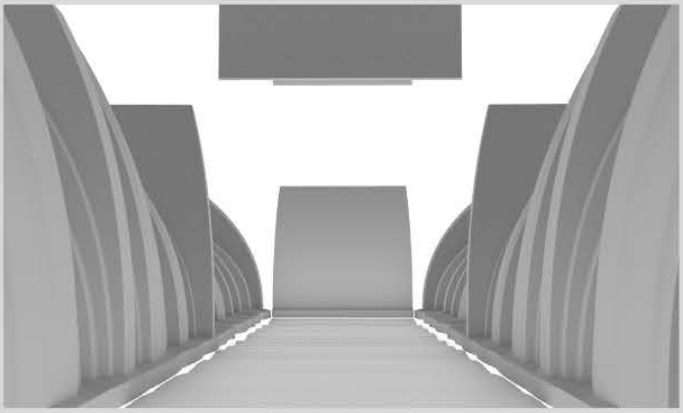
In the previous phase (Part 02), we verified that AI processes human emotion not as subjective feelings, but as quantifiable probability vectors.

Having established this data comprehension, this phase attempts to physicalize these variables. The objective is to map the extracted emotional values directly onto architectural elements—Scale, Light, and Texture—creating an environment that reacts to biological signals.

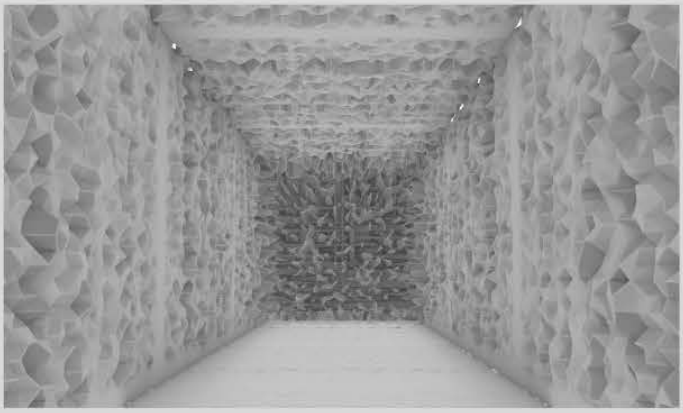
This experiment utilizes a Python script within Rhino 3D to translate raw EEG data directly into geometric deformation. Unlike conventional design where form follows function, here 'Form follows Data'. The corridor acts as a linear timeline, where the elements are physically affected by the intensity of the biological signal with AI generated rules. (by Gemini 3 pro)

ALGORITHMIC DIVERGENCE

While sharing the same generative framework, the algorithm branches into two distinct topological strategies based on the polarity of the input signal.



TYPE A : POSITIVE (Joy/Elation)
Operation: Addition & Inflation
Math: Pt + (Normal * Intensity * Smooth)
Result: The geometry expands outward. Continuous curvature creates a voluminous, airy atmosphere, symbolizing emotional liberation.



TYPE B : NEGATIVE (Anxiety/Sorrow)
Operation: Subtraction & Erosion
Math: Pt - (Normal * Intensity * Noise)
Result: The geometry collapses inward. High-frequency noise creates a rough, corroded texture, symbolizing psychological instability

WORKFLOW

STEP 1: INPUT

- EEG Dataset: extracts specific frequency bands to serve as the seed for spatial generation

STEP 2: SCRIPT

- Python: The script reads the EEG float values and maps them to geometric parameters.

STEP 3: GEOMETRY

- Rhino8: Physical displacement of the corridor mesh, divide into Wall, Ceiling, Floor Elements.

STEP 4: AI RENDER

- Z-image turbo: ComfyUI generates photorealistic textures Using ControlNet

STEP 5: COMBINE

- Photoshop: Combine rendered images with seperate elements from ComfyUI

OUTPUTS

POSITIVE EMOTION SPACE

PROMPTS (FLOOR)

(Masterpiece, best quality, 8k, photorealistic:1.2), bright and airy atmosphere, ethereal lighting, volumetric fog, cinematic lighting, futuristic organic architecture, Zaha Hadid style, white and cyan color palette, soft shadows, 16mm film grain.

wet surface look, pure reflection, mirror-like texture, reflecting the sky and clouds, clean and pristine, bright floor, minimal joint lines.

NEGATIVE EMOTION SPACE

PROMPTS (CEILING)

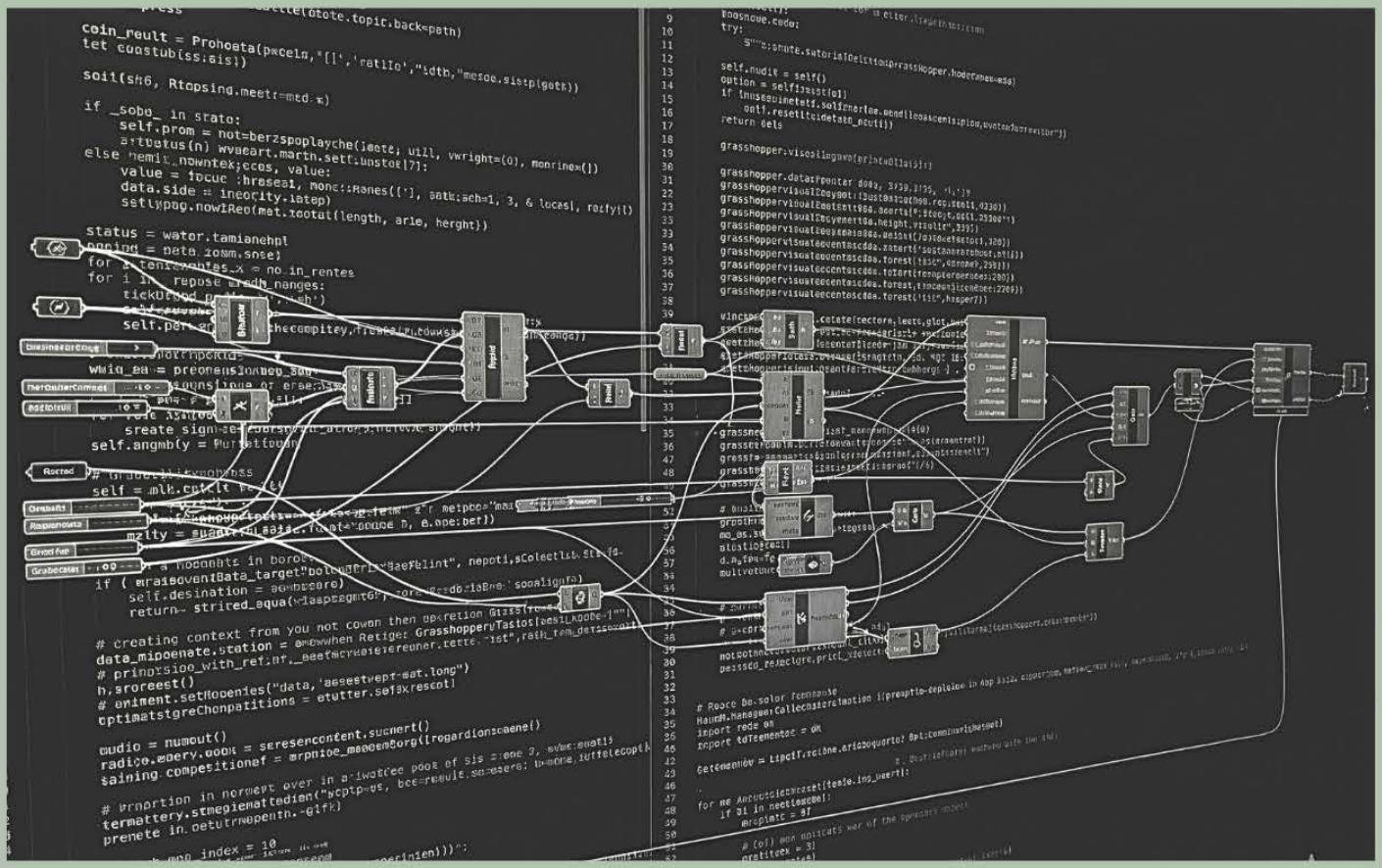
cinematic shot, photorealistic, highly detailed, oppressive horror atmosphere, dim volumetric fog, low key lighting, claustrophobic, 8k resolution, unreal engine 5 render style

suffocating ceiling, heavy wet concrete dripping with black sludge, moldy industrial pipes leaking dark fluid, stalactites of concrete hanging down, oppressive atmosphere, dim volumetric fog, shadowy void, high contrast

---

BUT THE QUESTION IS,
DOES THIS EVOKE ANY EMOTION?

CONCLUSION



THE EMOTION BEFORE LOGIC

This project proves that data is a building material. By bridging Python logic and Generative AI, I translated invisible biological signals—anxiety and joy—into tangible physical forms.

We are transitioning from an era of simply using AI to an era of living with AI. Architecture is no longer the exclusive domain of the human architect, and the spaces we inhabit will no longer be conceived solely by human hands.

The purpose of this research was to gauge AI's capacity to understand human affect. Ultimately, as we navigate through this rigid mechanical logic, we must move forward guided by the one thing that remains uniquely ours: Human Emotion.

EMOTIONAL ARCHITECTURE

JIHOON PARK

jhp503@gmail.com
+82 10 4096 5031

ADD+